GameZone Sales Analysis: Data Cleaning Project

1. Project Background & Business Overview

GameZone is a global e-commerce company (started in 2018) that sells new and refurbished gaming hardware through its website and mobile app. They have a solid amount of data on sales, marketing, and shipping, but it hasn't been fully used for business decisions yet.

I took on this project to clean and structure over 20,000 raw transaction records. The main goal here is to hand over a clean, reliable dataset to the business team so they can look at overall sales trends across different global regions without getting incorrect results because of messy data.

What the Business Cares About:

Try to understand overall sales trends across a variety of different regions.

2. Understanding the Data Structure

Before writing any cleaning formulas, I spent time understanding the "story" behind each row in the dataset:

- **The Grain:** Every single row represents a **unique customer order** identified by an `Order ID`.
- **The Metric:** The primary number we care about is `USD Price` (this is our revenue).
- **The Dimensions:** We have time details (`Purchase_Timestamp`, `Ship_Timestamp`), product names, platforms used, marketing channels, and country codes.

Story of one example record:

USER_ID	ORDER_ID	PURCHASE_TS	SHIP_TS	REFUND_TS	PRODUCT_NAME	PRODUCT_	USD_PRICE	PURCHASE_PLATFORM	MARKETING_CHANNE	ACCOUNT_CREATI	COUNTRY_CODE
2c06175e	0001328c3c220830	24/12/2020	26/12/2020		Nintendo Switch	e682	168	website	affiliate	unknown	US
ee8e5bc2	0002af7a5c6100772	01/10/2020	21/09/2020		Nintendo Switch	e682	160.61	website	direct	desktop	DE
9eb4efe0	0002b8350e167074	21/04/2020	16/02/2020		Nintendo Switch	8d0d	151.2	website	direct	desktop	US

"User XYZ placed an order on Dec 24th 2020 that got shipped out two days later.

It was a Nintendo Switch that was bought for almost $170 and it was purchased on the website.

And the user came to Gamezone through affiliate Marketing and we don't actually know how their account was created but they are located in the United States."

I followed two strict rules to keep a proper paper trail:

1. Never Touch the Raw Data.
2. Use Column Versioning.

3. Executive Data Quality Summary

First Pass at Data Cleaning:

1. Eyeball for obvious issues.
2. Filter to check distinct values.
3. Document in the Issue Log.

After running a quick sanity check across the columns, the overall data quality is actually quite good. Key fields like revenue and geography are mostly complete. However, I found a major logic issue in the shipping dates that affects about 10% of the entire dataset.

As an analyst, my instinct wasn't to just delete these rows. Deleting 2,000 orders just to make the data look "perfect" would wipe out a huge chunk of valid sales data and ruin our revenue reports. Instead, I fixed the minor typos programmatically and documented the bigger logic errors so the other team could look into the root cause.

4. Data Quality Audit & Issues Log

Here is the exact log of the issues I found, how big they were, and the logic behind my decisions:

Table	Column	Issue	Row_count	Magnitude	Solvable?	Resolution
orders	PURCHASE_TS	inconsistent date formate	10	0.05%	Y	used DATE function to reformate the dates correctly
orders	PURCHASE_TS	missing date	1	0.00%	N	left as is -low magnitude and no way to infer
orders	PRODUCT_NAME	inconsistent spelling between "27inches 4k gaming monitor" and "27in 4K gaming monitor"	61	0.28%	Y	recategorized to correct spelling
orders	USD_PRICE	missing or $0 transactions	34	0.16%	N	left as is - no way to infer, need to check with stakeholders
orders	MARKETING_CHANNEL	missing marketing channel	83	0.38%	Y	recategorized to "unknown"
orders	ACCOUNT_CREATION_METHOD	missing account creation method- same count as marketing channel?	83	0.38%	Y	recategorized to "unknown"
orders	COUNTRY_CODE	missing country code	37	0.17%	N	left as is - no way to infer, need to check with stakeholders
region	REGION	inconsistent or nonsensical values, missing values	9	0.04%	Y	filled in by looking up the correct values
orders	all	duplicates within order id for two different user IDs	145	0.66%	Y	left as is- could delete duplicate values but low volume
orders	all	ship timestamp before purchase timestamp	1999	9.14%	N	left it as is- need to flag if furthur analysis is related to ship timestamp

5. Data Augmentation(Adding Value)

Once the data was clean, I started thinking “How can I make my data more flexible & robust?”
In this case,

- I could slice my Purchase_timestamp into weeks, into months and years. Or could do the same for Shipment_timestamp but since time is not our key dimension, so not going to focus on that.
- I could calculate Time_To_Ship, calculating timestamp can be helpful in days knowing that we have some records where shipment date is before purchase date, it will come up as negative.
- I could bring in the Region to my lookup table, so I can see for every single purchase what region it was actually purchased in?
- If I had a user and customer table to reference then maybe I could pull in more demographics info like their street address, state or other info collected when they Sign up for an account.

6. Future Scope (How I'd scale this)

Data in the real world is never 100% perfect. If I were managing this data pipeline on the job, here is how I would approach scaling this process:

1. **Fix the Source:** The 10% shipping date error is clearly a system bug. I would sync with the software developers to add front-end validation on the website so a shipping date can never be entered as earlier than a purchase date.
2. **Automate Data Checks:** Instead of checking for duplicates or nulls manually in Excel or via ad-hoc scripts, I would set up automated data quality tests (using tools like **dbt** or Python scripts) to automatically flag these rows the moment new data enters the system.

My Full Data Cleaning Process straight from my head:

I understood the dataset by identifying the grain, metrics & dimensions and making sure that I can tell the story of an example record in the data.
I located solvable issues by conducting a first pass of cleaning and dealing with what I can in that process and then evaluated the unsolvable issues where I documented what I was finding and referred to another team if possible otherwise just leave it as is.
Then I started augmenting the data by what other info is available to me that I could use to make the dataset even more interesting.
And then finally, I noted and documented everything in the issue log.

One thing I have learned here is that data cleaning is like peeling onions, this is just the basics I have done here. More EDA and Data Cleaning is coming up in future.